Signature Features (Phase D design)

Identity: Think strategically. Master the siege. Smash the fortress. Conquer the kingdom.

Every level is a problem to solve (which hero, which material, which mechanism to trigger first) and every good solution should end in a big, readable collapse. This doc says what each pillar means in this game, what already exists, and what Phase D adds. Numbers go in `data/*.json`; the values here are starting points.

1. Material-specific interactions

Each hero has a `material_mult` per material (`data/projectiles.json`), so the right hero matters.

Material	Behaviour	Best answer	Status
Wood, crate, straw	Burns: fire spreads to touching wood	Blaze	done
Stone	Cracks in 3 stages, heavy	Rocky, Quaky	done
Powder barrel	Explodes, chains into other barrels	anything	done
Crystal	Shatters and cuts neighbours	Rocky	done
Ice (new)	Slippery (friction 0.05), low HP, shatters into shards; fire melts it fast (×3)	Blaze, Rocky	done
Iron (changed)	Barely damaged; a hard hit **bends** a beam into a V so pieces slide off it; fire makes it **red-hot** for 5 s (takes ×2.5 damage)	Blaze, then Rocky or Boomer	done
Rune crystal (new)	Magic: **absorbs** blasts, fire and quakes without damage, then **redirects** the energy as a beam in the direction of its rune arrow, breaking what is in line	aim a blast at it	done

The loadout screen shows the fortress materials and marks the heroes that are strong against them.

Built (numbers in `data/materials.json`): - **Ice** (`ice_block`, `ice_slab`, `ice_pillar`): friction 0.05, 40 HP, takes ×1.3. Fire melts it at 32 HP/s (no flames, no spreading) and Blaze does ×3. It shatters into ice shards that hurt blocks within 100 px. - **Iron bend:** a single hit of 14% of max HP or more folds an `iron_beam` into `iron_beam_bent` (a V with a two-part collider, +300 points, a chain link). It stays bent. - **Red-hot iron:** fire heats iron for 5 s instead of burning it. It glows orange and takes ×2.5. - **Rune crystal** (`rune_crystal`, decor `dir` in degrees for the arrow): 150 HP. Blasts, fire, quakes and beams are absorbed. After 0.35 s it fires a 1200 px beam (110 damage, per-material multipliers, ignites wood, heats iron, melts ice, 30 damage to guards, cuts ropes, snaps bolts). A beam that hits another rune crystal charges it, so beams can be bounced. 1 s cooldown. - **Loadout:** a FORTRESS strip lists materials with counts; tapping one shows its tip. Heroes whose block-weighted multiplier is at least ×1.2 get a STRONG HERE badge.

2. Interactive mechanical castles

Done: gates and drawbridges with pulley and counterweight rigs, moving platforms, swings, spikes, lava, cuttable rope loads, hanging platforms, crushers, geysers, bolted joints, weak points.

Phase D (built, tunables in `balance.json` → `lever`, `rotor`, `rune_switch`, `trapdoor`): - **Lever switch:** a shot, a fast block or a blast flips it and fires its `triggers` (open a gate, drop a crusher, open a trap door, stop a shield). 1,200 points. - **Windmill / rotating tower:** a hub with 2 to 4 sails turning at `dps`. Sails swat shots (and blocks with `hits_blocks`); a hit on the hub jams it

(2,500 points) and fires its `triggers`. - **Rotating shield:** iron plates orbiting a tower or core on their own physics layer, so they stop shots but never push the fortress. Time the gap, or hit a lever to stop the ring. - **Hidden mechanisms:** a rune switch under a block. It stays dark while covered; when the cover breaks it glows, shows "?", and a hit (3,000 points) springs a trap door under the archers. The flyover marks it with "?". All five are chain links (`mechanism`) and are tested in `_mechanism_d_tests` (11 checks).

The campaign set-piece pass also adds a flood gate and water simulation, a brake-triggered water wheel, a collapsing two-rope bridge, breakaway cliff ledges, a visible pulley/counterweight rig, a heavy throwing beam and an explosive cart. They appear in ordinary levels 12, 18, 19, 21, 22, 32 and 38, with crushing rope loads demonstrated in level 2. `--run-tests=contraptions` exercises each trigger and resulting collapse.

3. Spectacular chain-reaction destruction

Status: **done** (`src/game/chain.gd`, `src/game/replay.gd`, numbers in `balance.json` → `chain`, `replay`). - **Chain:** every break, KO, barrel blast, rope cut, bolt snap and trigger within 1.2 s of the last one adds 1 to the chain. Multiplier = 1 + 0.1 per link after the first, up to ×2.5. When the chain ends, a chain of 3+ links pays its points again × (multiplier – 1). - **HUD:** a chain chip under the score (from 3 links) pops on each link and heats from gold to red. Milestones float in the world and flash the chip: 5 "NICE!", 10 "GREAT CHAIN!", 20 "SMASHING!", 35 "LEGENDARY!". - **Cinematic camera:** from 8 links the camera eases toward the centre of the chain and zooms in a little. At 12 links, or a barrel blast in a chain of 4+, time slows to 0.35× for 0.8 s (4 s cooldown). - **Replay:** the level records every block, archer, hero and moving part at 20 Hz (up to 90 s). The result screen's REPLAY button hides the HUD and plays the best chain (1.5 s either side) at 0.5× with letterbox bars. Tap to skip. - Results show blocks smashed, guards defeated, heroes saved, best chain and chain bonus. The 3-star score expects some chain bonus (`gen_levels.py`: base + 0.75 × block score + 10,000).

4. Multi-stage boss castles

Done: crowned boss, HP phases (reinforce, shield, enrage, charge), moving castles, boss bar.

Phase D turns each boss fight into stages: **disable the defences, expose the core, finish the boss.**

Status: built (`src/game/boss_rig.gd`, tunables in `balance.json` → `boss_rig`, materials `core` and `cannon`, 14 checks in `_boss_rig_tests`). Breaking the last core scores 5,000, drops the shield, stuns the boss for 3 s and plays a slow-motion beat. Cannonballs ignore soldier shields and hit twice. Engineers rebuild a broken block within 700 px every 6 s (2,000 points each). - **Power core:** a glowing core block. While any core stands, the boss has an unbreakable shield. - **Rotating armor:** iron plates orbit the core (pillar 2 rotating shield). - **Defensive weapons:** cannon turrets fire cannonballs at your soldiers (2 damage); they are iron targets you can destroy. - **Repair system:** engineers walk to broken spots and rebuild a destroyed block every 6 s until you knock them out. - The boss bar shows status icons: core (lit / broken), armor, turrets, engineers.

Boss	Castle	Defences	Core
L10 Siege Warlord	siege tower on wheels	1 cannon, 1 engineer	war drum in the base
L20 Sandstone Sultan	citadel with sliding keep	rotating stone armor	gem in the keep
L30 Captain Ironhook	iron galleon on the tide	2 cannons, 2 engineers	powder magazine (explodes)
L40 The Forgemaster	foundry on rails	rotating iron armor, engineer, lava geyser	furnace
L50 The Iron Baron	airship in the wind	rotating armor, 2 cannons, 2 engineers	2 balloon cores

5. Game modes and objectives

A level's `mode` sets its rules. Campaign levels mix them for variety; the **Challenges** screen (opens after level 10) has the rest. Each mode has its own HUD objective chip.

Mode	Goal	Stars by
Siege (default)	Defeat every archer	score
Rescue (done)	Free the captives and defeat the archers	score
Treasure heist	Break open the chests. A chest that falls into the gap is lost	chests + score
Time attack	Clear the fortress before the timer runs out; heroes keep coming	time left
One shot	One hero. Bring it all down with one chain reaction	score
Survival siege	Reinforcements arrive in waves by raft or airship; survive N waves	waves
Reverse siege	Enemy siege towers and ships cross the gap toward you; sink each one before it docks	engines sunk
Boss rush	All 5 bosses back to back, shared heroes, soldier HP carries over (after L50)	time

A **daily challenge** picks one challenge by date and pays 100 coins once per day.

Status: built. `ModeRules` (`src/game/mode_rules.gd`) runs the rules from level JSON `mode` and `mode_cfg`; tunables in `balance.json` → `modes`. Eight challenge levels (ids 101–108, authored in `tools/campaign.py` → `CHALLENGES`, written to `data/challenges/`): Gold Rush and Pirate Hoard (heist), Beat the Clock and Forge Sprint (time attack, 40 and 50 s), One Perfect Shot (one Boomer), Hold the Line (3 survival waves), Sky Raiders and Harbor Assault (reverse siege). Boss Rush runs levels 10–50 with an 18-hero shared pool after level 50 (stars at ≤ 450 / 330 / 240 s). The Challenges screen opens after level 10 (map button); the daily card is first with a countdown. 28 checks in `_mode_tests`.

Master Sieges: ten hard fixed-four-shot mechanic puzzles (ids 121–130) appear on the same screen. The pack unlocks at 25 campaign stars. Before that, one locked card shows live star progress; after that, all ten violet-gold cards appear. Its separate `specials` index keeps it out of daily rotation. Every level declares `required_mechanics`, covering flood, water wheel/gears, collapsing bridge, breakable cliff, pulley/counterweight, explosive cart and heavy seesaw launch. The focused `--run-tests=specials` suite validates the gate, catalog, mechanics, stability and phone framing.

6. Kingdom conquest and progression

Status: logic built (`src/game/progression.gd`, 20 checks in `_progression_tests`; screens in task 8). Data: `balance.json` → `siege_weapons`, `upgrades` (Vanguard armor +1 HP ×2, shields +8% ×3, weapon power +4% ×3, reload –8% ×3; later levels cost blueprints), `cosmetics` (8 skins, 5 trails; tints and trail colour only), `realm_chests` (10/20/30 stars), `rank`, `secrets` (scrolls in levels 7, 15, 27, 36, 46 unlock secret levels 111–115), `achievements` (20). The loadout screen has a siege-weapon chip (tap to cycle unlocked weapons).

- **Regions:** 8 realms with their own gap, hazards, castle style and boss (done). Clearing a realm boss flips its map castle to our blue flag.
- **Siege weapons** (pick on the loadout screen):
- Catapult: balanced.

- Ballista (unlocks after L20): flat, fast shots for front walls and turrets.
- Trebuchet (unlocks after L30): high lobs with +35% hero mass for enclosed keeps; reloads slower.
- **Upgrades:** the Wizard's Workshop (heroes, done) gains Vanguard tracks (soldier HP, shield chance) and siege weapon tracks (reload, power).
- **Cosmetics:** catapult skins (Sakura, Desert Gold, Pirate, Forge, Sky) and hero trails (petals, sparks, stars). Earned from realm star chests, achievements or coins. Never stronger, only different.
- **Secret levels:** one per realm. A hidden map scroll sits in a crate in one level of the realm. Hit it to reveal a dotted path to a secret node on the map.
- **Realm chests** on the map at 10, 20 and 30 stars per realm.
- **Kingdom rank** from total stars: Squire, Knight, Captain, Baron, Duke, King.
- **Achievements:** about 20, for example "Chain of 20", "Bend 10 iron beams", "Beat a boss with the core first", "Sink 5 siege ships".

Phase D build order

1. Phase C leftovers: cutout support for mechanism/block pieces; APK size pass.
2. Chain reactions (chain, HUD, cinematic camera, slow-mo, replay, results breakdown).
3. Materials (ice, iron bend and heat, rune crystal, loadout material panel).
4. Mechanisms (lever, windmill, rotating shield, hidden switch and trap door).
5. Boss stages (core, rotating armor, cannon turrets, engineers) and the 5 boss redesigns.
6. Modes (heist, time attack, one shot, survival, reverse, boss rush), challenge levels, daily.
7. Progression (siege weapons, Vanguard upgrades, cosmetics, secret levels, realm chests, rank, achievements).
8. Screens (see `UI_DESIGN.md`): icon set, home, map, HUD chips, results, pause, settings, armory, challenges, achievements, daily reward.
9. Weave the new content into the campaign, tests, docs, AI prompts, APK.